

Project 3: Corporate Default Prediction

Altman-Style Scores, Logistic Regression, and Machine Learning

Financial Analytics / Credit Risk Modeling

1 Purpose of the Project

This project asks you to build predictive models of corporate bankruptcy using actual U.S. public-company financial data. The exercise connects three layers of credit-risk modeling: classical accounting-based distress scores, interpretable statistical models, and modern machine-learning methods.

The practical question is simple:

Using financial-statement information available today, can we identify firms that are more likely to enter bankruptcy in the future?

Every team will complete the full analysis. Each team will submit its own report, code, tables, figures, and model results. The class presentation will be handled differently: **instead of seven similar team presentations, representatives from the seven teams will produce one coordinated high-level presentation that summarizes what the class learned from the project.**

2 Dataset

We will use the **U.S. Company Bankruptcy Prediction Dataset**, available on Kaggle and originally associated with the American public-company bankruptcy dataset documented by Lombardo et al. The dataset contains accounting data for U.S. public companies listed on NYSE and NASDAQ over the period 1999–2018. The original dataset documentation reports 8,262 companies observed over 78,682 firm-year observations. A firm is labeled bankrupt if it files under Chapter 11 or Chapter 7 of the U.S. Bankruptcy Code in the subsequent fiscal year.

The prediction problem should be treated as a forward-looking task. Financial variables from fiscal year t are used to predict bankruptcy after fiscal year t :

$$Y_{i,t+1} = \begin{cases} 1, & \text{if firm } i \text{ enters bankruptcy after fiscal year } t, \\ 0, & \text{otherwise.} \end{cases}$$

A random train-test split is not the preferred design because it mixes time periods. A realistic credit-risk model should be estimated on earlier years and evaluated on later years. Unless your team has a strong reason to do otherwise, use the following sample split:

Sample	Years	Use	Purpose
Training	1999–2011	Model estimation	Estimate Altman-style, logistic, and machine-learning models.
Validation	2012–2014	Model choice	Select variables, tune hyperparameters, choose thresholds.
Test	2015–2018	Final evaluation	Report final out-of-sample performance.

3 Required Team Analysis

Each team must complete all parts below. Do not divide the project so that one team only does Altman, another only does logistic regression, and another only does machine learning. Every team is responsible for the complete modeling workflow.

3.1 Data Preparation and Financial Understanding

Begin by understanding the data before running models. At minimum, report:

- the number of firm-year observations;
- the number and percentage of bankrupt observations;
- the bankruptcy rate by sample period;
- whether the data contain missing values or extreme observations;
- the financial meaning of the main accounting variables used in your models.

Bankruptcy is a rare event. This class imbalance should affect both your model evaluation and your discussion. A model that predicts “no bankruptcy” for every firm may look accurate, but it is not useful for credit-risk screening.

3.2 Altman-Style Accounting Benchmark

Construct an accounting-based distress score inspired by Altman’s bankruptcy-prediction framework. The original Altman Z-score combines ratios related to liquidity, cumulative profitability, current profitability, leverage, and asset productivity [1]. The exact variables in this dataset may not match the original model perfectly, so your job is to build the closest defensible version using available variables.

Your Altman-style section should include:

1. a table showing the ratios you constructed and their financial interpretation;
2. a clear explanation of whether each ratio should increase or decrease default risk;
3. a simple scoring rule, index, or standardized weighted score;
4. an evaluation of whether the score separates future bankrupt firms from non-bankrupt firms.

The point of this section is not to mechanically reproduce a historical formula. The point is to create a transparent benchmark that a credit analyst could understand.

3.3 Logistic Regression

Estimate at least one logistic regression model. The baseline model should use financially motivated variables rather than an indiscriminate list of all columns. A standard specification is

$$\Pr(Y_{i,t+1} = 1 \mid X_{i,t}) = \frac{1}{1 + \exp[-(\beta_0 + \beta' X_{i,t})]}.$$

Your logistic regression section should discuss:

- the economic interpretation of the main coefficient signs;
- whether leverage, profitability, liquidity, and size behave as expected;
- whether logistic regression improves on the Altman-style benchmark;
- whether the model remains interpretable after adding more variables.

You may estimate additional versions, such as LASSO logistic regression or a model with industry/year controls, but a clean baseline model must be included.

3.4 Machine-Learning Models

Estimate at least two machine-learning models. Acceptable choices include classification trees, random forests, gradient boosting, XGBoost, support-vector machines, or neural networks. Tree-based models are especially natural for this project because they can capture nonlinear risk thresholds and interactions among accounting variables.

Your machine-learning section should include:

- a short explanation of each model used;
- the tuning process used on the validation sample;
- final test-sample performance;
- variable-importance or model-explanation results;
- a comparison with the logistic regression and Altman-style benchmark.

Do not report machine-learning output as a black box. The report should still sound like a credit-risk analysis, not only a coding exercise.

4 Model Evaluation

Because bankruptcies are rare, overall accuracy should not be your main metric. Each team must report a set of metrics that reflects both statistical performance and credit-screening usefulness.

Metric	Why it matters
ROC-AUC	Measures whether the model ranks bankrupt firms above non-bankrupt firms.
PR-AUC	More informative when the positive class is rare.
Recall / sensitivity	Shows how many actual bankruptcies the model captures.
Precision	Shows how many predicted high-risk firms actually go bankrupt.
Confusion matrix	Shows the tradeoff between false positives and false negatives at a chosen threshold.
Top-risk-bucket recall	Measures how many bankruptcies are captured among the riskiest 5% or 10% of firms.

A useful credit-risk model does not have to classify every firm correctly. In practice, it is often used to prioritize attention. For that reason, each team must include a chart or table showing how many future bankruptcies appear in the highest predicted-risk buckets.

5 Required Deliverables

Each team submits a complete project package.

5.1 Full Team Report

Submit a report of approximately 8–12 pages, excluding appendix material. The report should include:

1. data description and sample split;
2. bankruptcy-rate and class-imbalance discussion;
3. Altman-style benchmark construction and results;
4. logistic regression specification and interpretation;
5. machine-learning model design and tuning;
6. model-comparison tables;
7. top-risk-bucket analysis;
8. final credit-risk interpretation;
9. limitations of the analysis.

The report should be written as an applied financial analytics report. Explain your modeling choices in plain financial language. Tables and figures should support the argument, not simply document software output.

5.2 Code File

Submit a clean R or Python file. The code should run from beginning to end and reproduce the main results in the report. It should load the data, prepare variables, create the sample split, estimate models, generate evaluation metrics, and produce the main tables and figures.

5.3 Model Prediction File

Submit a file containing your final predicted bankruptcy probabilities for the test sample. This file will be used to construct the class-level model comparison table.

6 One Joint Class Presentation

There will be one class presentation, not seven separate team presentations. Each team still completes the full report, but the oral presentation will be organized as a single coordinated briefing by representatives from all seven teams.

The purpose is to avoid repetition and to make the final session more useful. We do not need to hear seven versions of the same sequence: dataset, logistic regression, random forest, XGBoost, AUC. Instead, we want one clear synthesis of what the class learned from doing the full exercise independently.

Each team will appoint one representative. The seven representatives will meet before the presentation and assemble one common slide deck. The presentation should not be a stitched-together set of mini-presentations. It should sound like one analyst briefing prepared by a credit-risk modeling group.

6.1 Suggested Presentation Roles

Rep.	Role	Presentation responsibility
1	Framing	Explain the prediction problem, bankruptcy definition, and why this is a forward-looking credit-risk exercise.
2	Data	Summarize the dataset, sample split, bankruptcy rate, and class imbalance.
3	Accounting benchmark	Present how the class translated Altman-style logic into ratios available in the dataset.
4	Logistic model	Summarize the main logistic regression patterns: signs, interpretation, and benchmark performance.
5	Machine learning	Summarize which ML methods were tried and where they improved prediction.
6	Model comparison	Present the common leaderboard: ROC-AUC, PR-AUC, recall in top-risk buckets, and model stability.
7	Credit interpretation	Explain what the results imply for a lender, investor, credit analyst, or risk manager.

Each representative should use results from all teams where useful, not only the results from their own team. For example, the model-comparison representative should summarize the range of performance across teams, not only one team’s best model.

6.2 Suggested Slide Structure

The common presentation should contain approximately 15–20 slides:

Section	Slides	Content
Problem and design	2–3	What is being predicted, why timing matters, what data are used.
Class imbalance	2	Bankruptcy rate, why accuracy is misleading, evaluation logic.
Altman-style benchmark	2–3	Ratios used, financial interpretation, benchmark performance.
Logistic regression	2–3	Coefficient signs, selected variables, predictive performance.
Machine learning	3–4	Models used, tuning, performance, variable importance.
Class leaderboard	2	Team-level comparison of best submitted models.
Credit interpretation	2–3	What the models reveal, when they are useful, what they miss.

6.3 Class-Level Leaderboard

The class presentation should include a model-comparison table. The ranking should not be based on accuracy. Use metrics that make sense for rare-event prediction.

Team	Best model	ROC-AUC	PR-AUC	Top 10% recall	Interpretation note
Team 1					
Team 2					
Team 3					
Team 4					
Team 5					
Team 6					
Team 7					

The leaderboard is meant to create comparison, cross-learning, and discussion, not to turn the course project into a contest. A model with slightly lower predictive performance may be more useful if it is stable, interpretable, and financially sensible. Let us learn from each other.

7 References

References

- [1] Altman, E. I. (1968). Financial ratios, discriminant analysis and the prediction of corporate bankruptcy. *The Journal of Finance*, 23(4), 589–609.
- [2] Ohlson, J. A. (1980). Financial ratios and the probabilistic prediction of bankruptcy. *Journal of Accounting Research*, 18(1), 109–131.
- [3] Lombardo, G., De Luca, G., and colleagues. (2022). Machine learning for bankruptcy prediction in the American stock market. *Future Internet*, 14(8), 244.
- [4] Lombardo, G. et al. *Bankruptcy Prediction Dataset for American Companies in the Stock Market*. GitHub repository: https://github.com/sowide/bankruptcy_dataset
- [5] Utkarsh X. *US Company Bankruptcy Prediction Dataset: Accounting Data of NYSE and NASDAQ Companies, 1999–2018*. Kaggle dataset: <https://www.kaggle.com/datasets/utkarshx27/american-companies-bankruptcy-prediction-dataset>